

Week 7

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Student Performance: Data Cleaning & Transformation

Introduction

To clean, transform, and prepare the Student Performance dataset for analysis in Power BI.

Data Cleaning

The dataset was first inspected to identify the available columns, data types, missing values, errors, and duplicate records. Duplicate records were removed where appropriate, invalid or missing values were handled, and the data types were corrected. Column names were also reviewed for clarity.

The screenshot shows the Power BI Query Editor interface. The formula bar at the top contains the following DAX formula:

```
= Table.AddColumn(#"Added Conditional Column", "Result", each if [Average] >= 40 then "Pass" else "Fail")
```

The table below shows the data structure with columns: gender, race/ethnicity, parental level of education, lunch, test preparation course, and math score. Each column has a data type icon and a status indicator (Valid, Error, Empty) with a percentage.

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score
1	female	group B	bachelor's degree	standard	none	7.
2	female	group C	some college	standard	completed	6.
3	female	group B	master's degree	standard	none	9.
4	male	group A	associate's degree	free/reduced	none	4.
5	male	group C	some college	standard	none	7.
6	female	group B	associate's degree	standard	none	7.
7	female	group B	some college	standard	completed	8.
8	male	group B	some college	free/reduced	none	4.
9	male	group D	high school	free/reduced	completed	6.
10	female	group B	high school	free/reduced	none	3.
11	male	group C	associate's degree	standard	none	5.
12	male	group D	associate's degree	standard	none	4.
13	female	group B	high school	standard	none	6.
14	male	group A	some college	standard	completed	7.
15	female	group A	master's degree	standard	none	5.
16	female	group C	some high school	standard	none	6.
17	male	group C	high school	standard	none	8.
18	female	group B	some high school	free/reduced	none	1.
19	male	group C	master's degree	free/reduced	completed	4.
20	female	group C	associate's degree	free/reduced	none	5.
21	male	group D	high school	standard	none	6.
22	female	group B	some college	free/reduced	completed	6.
23	male	group D	some college	standard	none	4.
24	female	group C	some high school	standard	none	6.
25	male	group D	bachelor's degree	free/reduced	completed	7.

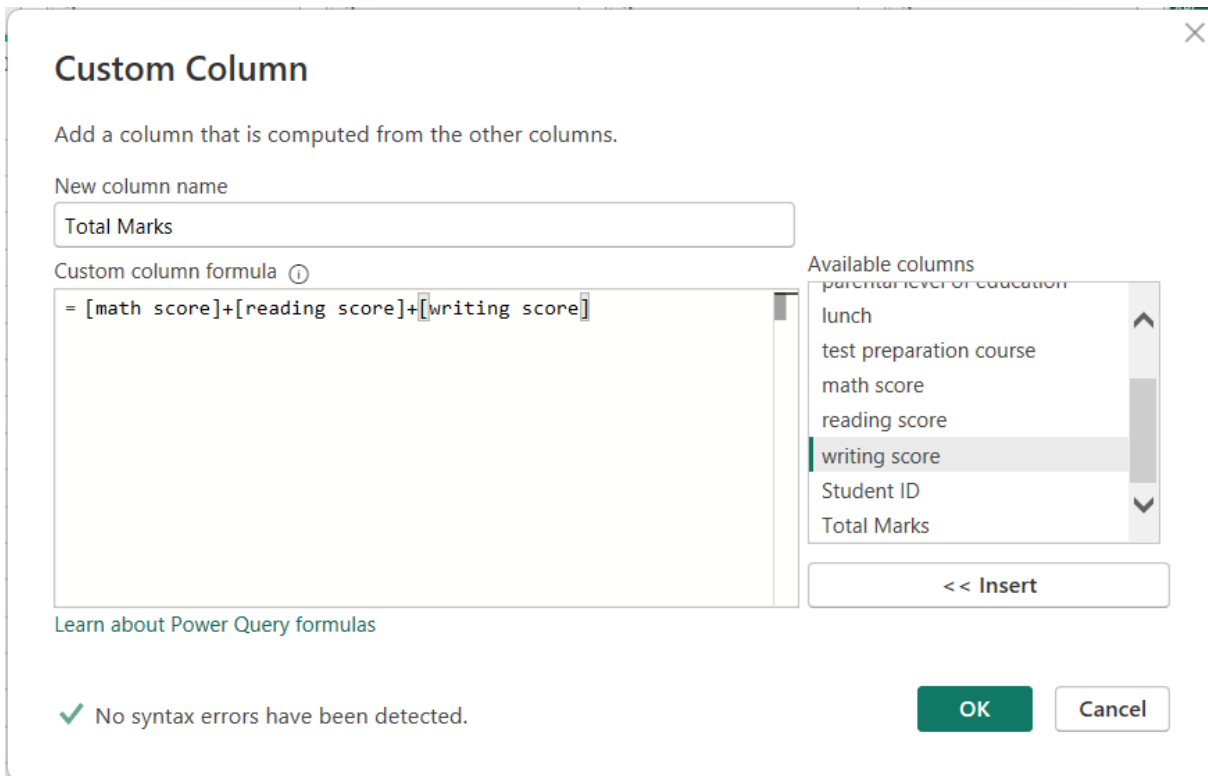
Calculated Columns

Calculated columns were created to measure student performance.

Total Marks

The total score was calculated using:

Math Score + Reading Score + Writing Score



Custom Column

Add a column that is computed from the other columns.

New column name
Total Marks

Custom column formula ⓘ
= [math score]+[reading score]+[writing score]

Available columns
parental level of education
lunch
test preparation course
math score
reading score
writing score
Student ID
Total Marks

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

Average Score

The average score was calculated using: Total Score / 3.

Custom Column

×

Add a column that is computed from the other columns.

New column name

Average

Custom column formula ⓘ

= [Total Marks]/3

Available columns

test preparation course

math score

reading score

writing score

Student ID

Total Marks

Average

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

Student Marks Table

The subject score columns were reshaped using Unpivot Columns. The Math Score, Reading Score, and Writing Score were converted into a structured format containing:

- Student ID
- Subject
- Score

This makes it possible to analyze student performance across different subjects.

Queries [4] < ✕ ✓ fx = Table.RenameColumns(#"Unpivoted Columns",{{"Attribute", "Subject"}, {"Va

	Student ID	Subject	Score
	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%
1	1	math score	72
2	1	reading score	72
3	1	writing score	74
4	2	math score	69
5	2	reading score	90
6	2	writing score	88
7	3	math score	90
8	3	reading score	95
9	3	writing score	93
10	4	math score	47
11	4	reading score	57
12	4	writing score	44
13	5	math score	76
14	5	reading score	78
15	5	writing score	75
16	6	math score	71
17	6	reading score	83
18	6	writing score	78
19	7	math score	88

DAX Measures

Average Math Score

Data Queries Relationships Calculations Parameters

```

1 Average Math Score =
2 CALCULATE(
3     AVERAGE('Student Marks'[Score]),
4     'Student Marks'[Subject] = "Math"
5 )

```

Students Details

- gender
- lunch
- parental level of education
- race/ethnicity
- Student ID
- test preparation course

Student Marks

- Score

MeasureTable

- Average Math Score

Average Reading Score

```

1 Average Reading Score =
2 CALCULATE(
3     AVERAGE('Student Marks'[Score]),
4     'Student Marks'[Subject] = "Reading"
5 )

```

The screenshot shows the Power BI Desktop interface. The DAX formula bar contains the following code:

```

1 Average Reading Score =
2 CALCULATE(
3     AVERAGE('Student Marks'[Score]),
4     'Student Marks'[Subject] = "Reading"
5 )

```

The 'Student Marks' table is visible in the Fields pane, with the 'Score' column selected. The 'Students Details' table is also visible, with columns: gender, lunch, parental level of education, race/ethnicity, Student ID, and test preparation course. The 'MeasureTable' is shown on the right, containing 'Average Math Score' and 'Average Reading Score'.

Average Writing Score

```

1 Average Writing Score =
2 CALCULATE(
3     AVERAGE('Student Marks'[Score]),
4     'Student Marks'[Subject] = "Writing"
5 )

```

The screenshot shows the Power BI Desktop interface. The DAX formula bar contains the following code:

```

1 Average Writing Score =
2 CALCULATE(
3     AVERAGE('Student Marks'[Score]),
4     'Student Marks'[Subject] = "Writing"
5 )

```

The 'Student Marks' table is visible in the Fields pane, with the 'Score' column selected. The 'Students Details' table is also visible, with columns: gender, lunch, parental level of education, race/ethnicity, Student ID, and test preparation course. The 'MeasureTable' is shown on the right, containing 'Average Math Score', 'Average Reading Score', and 'Average Writing Score'.

Overall Average Score

```

1 Overall Average Score =
2 AVERAGE('Student Marks'[Score])

```

The screenshot shows the Power BI Desktop interface. The DAX formula bar contains the following code:

```

1 Overall Average Score =
2 AVERAGE('Student Marks'[Score])

```

Overall Pass %

Overall Pass % =

```

1 DIVIDE(
2   CALCULATE(
3     COUNTROWS('StudentsPerformance'),
4     'StudentsPerformance'[Result] = "Pass"
5   ),
6   COUNTROWS('StudentsPerformance')
7 ) * 100

```

lunch
parental level of education
race/ethnicity
Student ID
test preparation course

Student Marks ...

Σ Score

MeasureTable ...

- Average Math Score
- Average Reading Score
- Average Writing Score
- Overall Average Score
- Overall Pass %**

Validation of Results

The calculated scores, grades, Pass/Fail results, relationships, and DAX measures were checked to ensure they were correct and ready for visualization.

